

File Edit Selection View Go Run Terminal Help python_test

EXPLORER

- PYTHON_TEST
 - .vscode
 - settings.json
 - New folder
 - black.jpg
 - create_test_image_day3.py
 - create_test_image.py
 - DAY-01 Program Outputs.pdf
 - DAY-02 Program outputs.pdf
 - DAY-03 Program Outputs.pdf
 - drawing_basics.py
 - exercise1.day3.py
 - exercise1.house.py**
 - exercise1.py
 - exercise2.day3.py
 - exercise2.py
 - exercise3.day3.py
 - exercise3.py
 - exercise4.day3.py
 - getting_and_setting.py
 - gradient.png
 - image.py
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 - pixel_basics.py
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 - terex.png
 - test_gradient.png
- OUTLINE
- TIMELINE

exercise1.house.py

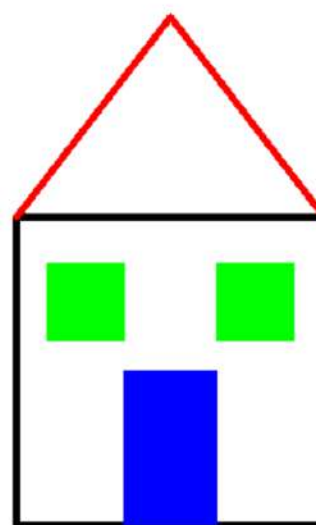
```
1 # exercise_1_house.py
2
3 import numpy as np
4 import cv2
5
6 # Create canvas
7 canvas = np.ones((500, 500, 3), dtype="uint8") * 255 # White background
8
9 # Draw the house body (rectangle)
10 cv2.rectangle(canvas, (150, 250), (350, 450), (0, 0, 0), 3) # Black outline
11
12 # Draw the roof (triangle using lines)
13 cv2.line(canvas, (150, 250), (250, 120), (0, 0, 255), 3) # Red line
14 cv2.line(canvas, (250, 120), (350, 250), (0, 0, 255), 3) # Red line
15
16 # Draw the door
17 cv2.rectangle(canvas, (220, 350), (280, 450), (255, 0, 0), -1) # Blue door
18
19 # Draw windows
20 cv2.rectangle(canvas, (170, 280), (220, 330), (0, 255, 0), -1) # Green window
21 cv2.rectangle(canvas, (270, 280), (320, 330), (0, 255, 0), -1) # Green window
```

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PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise1.house.py

House

My Dream House



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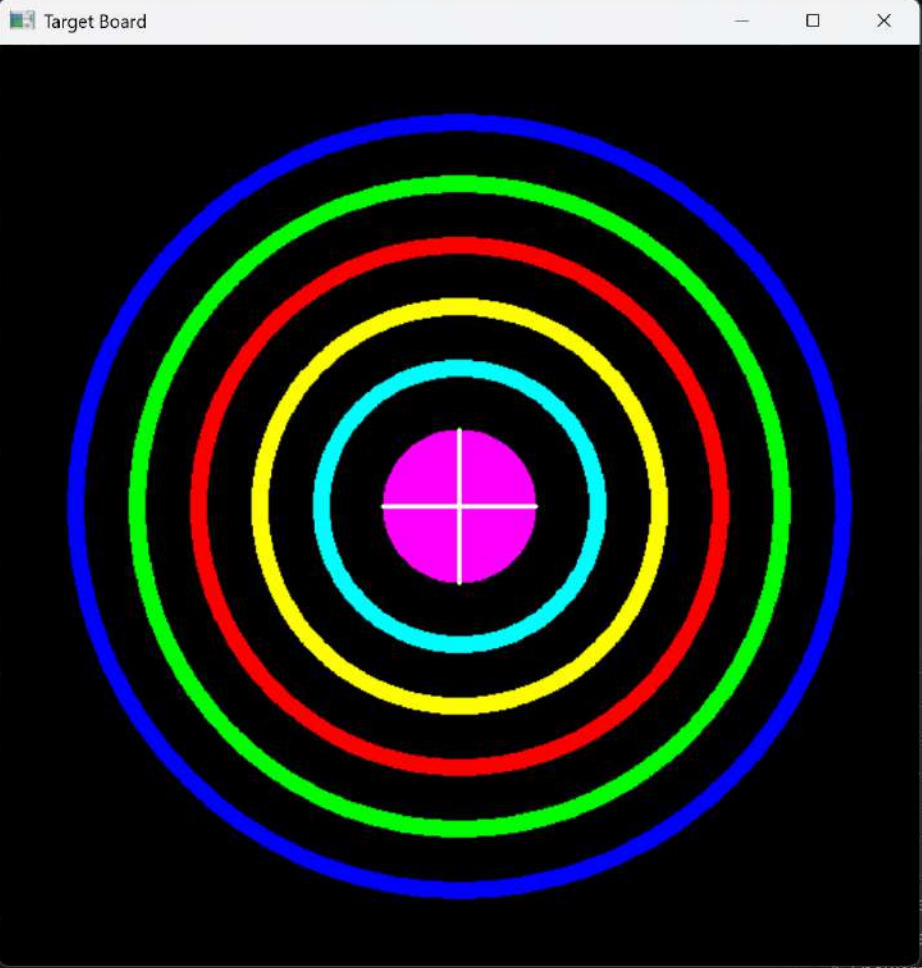
exercise2.target.py X

```
1 # exercise_2_target.py
2
3 import numpy as np
4 import cv2
5
6 WHITE = (255, 255, 255)
7
8 canvas = np.zeros((600, 600, 3), dtype="uint8")
9
10 center = (300, 300)
11
12 # Colors for target rings (outer to inner)
13 colors = [
14     (255, 0, 0),      # Blue
15     (0, 255, 0),      # Green
16     (0, 0, 255),      # Red
17     (0, 255, 255),    # Yellow
18     (255, 255, 0),    # Cyan
19     (255, 0, 255)     # Magenta
20 ]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exerci

Target Board



powerShell
powerShell
powerShell
powerShell
python

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 - red.jpg
 - saved_image.jpg
- OUTLINE
- TIMELINE

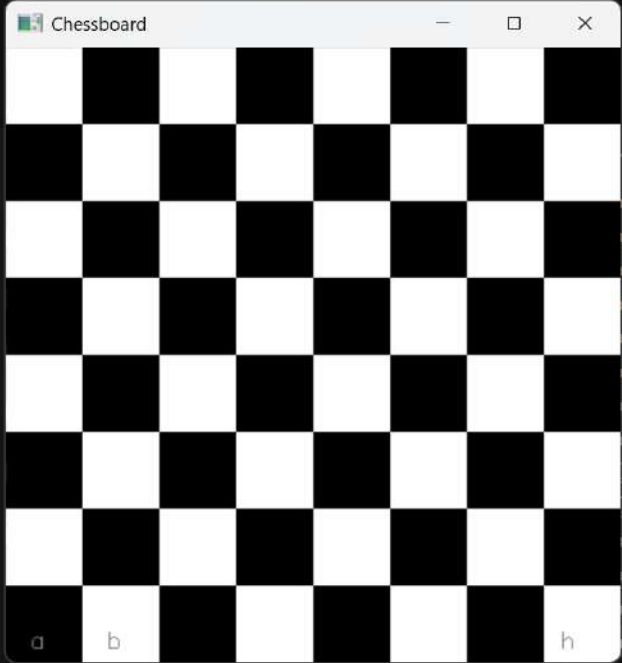
exercise3.chessboard.py X

```
1 # exercise_3_chessboard.py
2
3 import numpy as np
4 import cv2
5
6 # Define colors
7 WHITE = (255, 255, 255)
8 BLACK = (0, 0, 0)
9
10 # Create 8x8 chessboard on a 400x400 canvas
11 square_size = 50
12 board = np.zeros((400, 400, 3), dtype="uint8")
13
14 for row in range(8):
15     for col in range(8):
16
17         # Calculate pixel coordinates
18         y_start = row * square_size
19         y_end = (row + 1) * square_size
20         x_start = col * square_size
```

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PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise3.chessboard.py

Chessboard



Ln 58, Col 46 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)

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EXPLORER

PYTHON_TEST

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- DAY-03 Program Outputs.pdf
- drawing_basics.py
- exercise1.day3.py
- exercise1.house.py
- exercise1.py
- exercise2.day3.py
- exercise2.py
- exercise2.target.py
- exercise3.chessboard.py
- exercise3.day3.py
- exercise3.py
- exercise4.annotations.py**
- exercise4.day3.py
- getting_and_setting.py
- gradient.png
- house.png
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- load_display_save.py
- my_masterpiece.png
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- output.jpg
- output.tiff
- pixel_basics.py
- red.jpg
- saved_image.jpg
- saved_image.png
- target.png
- terex.png
- test_gradient.png
- test_image_gray.png
- test_image.png

exercise4.annotations.py X

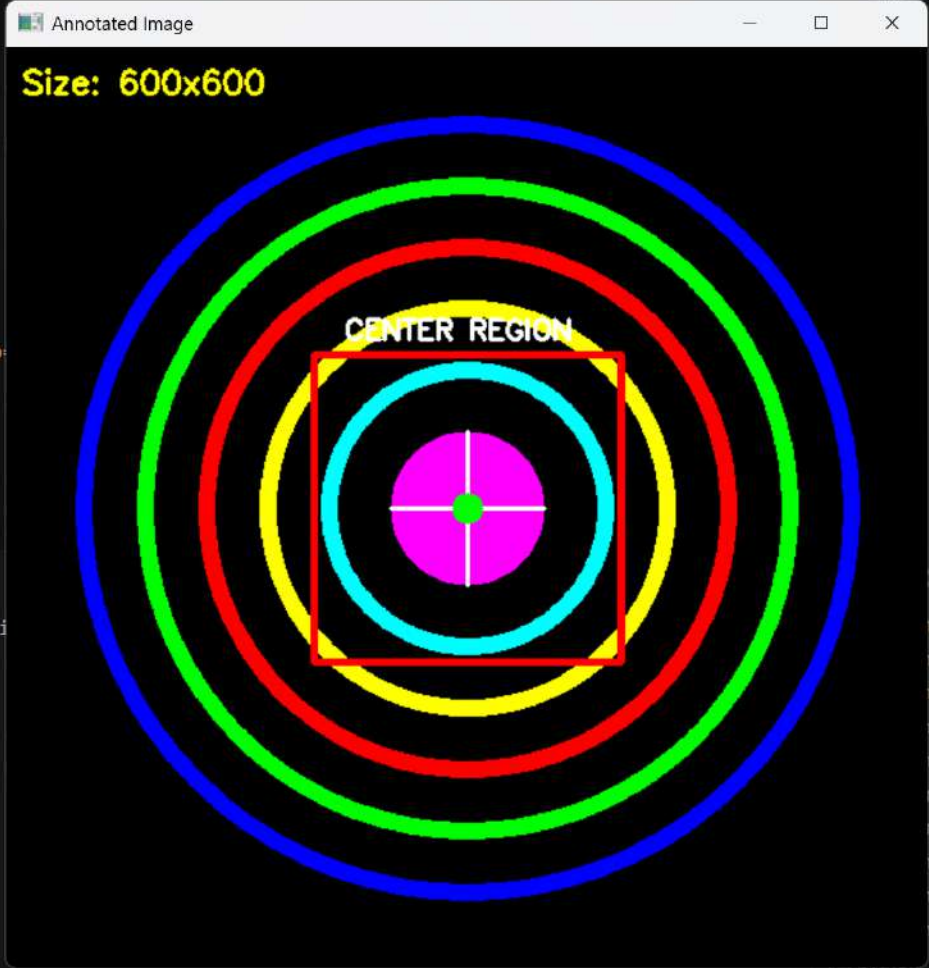
```
1 # exercise_4_annotate.py
2
3 import argparse
4 import cv2
5
6 # Define colors (BGR format)
7 RED = (0, 0, 255)
8 GREEN = (0, 255, 0)
9 WHITE = (255, 255, 255)
10 YELLOW = (0, 255, 255)
11
12 # Parse command-line arguments
13 ap = argparse.ArgumentParser()
14 ap.add_argument("-i", "--image", required=True, help=
15 args = vars(ap.parse_args())
16
17 # Load image
18 image = cv2.imread(args["image"])
19
20 if image is None:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exerci

Annotated Image

Size: 600x600



The image displays a target-like graphic with concentric circles. The circles are colored in a sequence: a central magenta circle, followed by a cyan ring, a yellow ring, a red ring, a green ring, and an outer blue ring. A red square is superimposed over the center, with the text "CENTER REGION" written in white above it. The entire graphic is set against a black background. The image is displayed in a window titled "Annotated Image" with a size of 600x600 pixels.

Ln 59, Col 2 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)

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- terex.png
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OUTLINE

TIMELINE

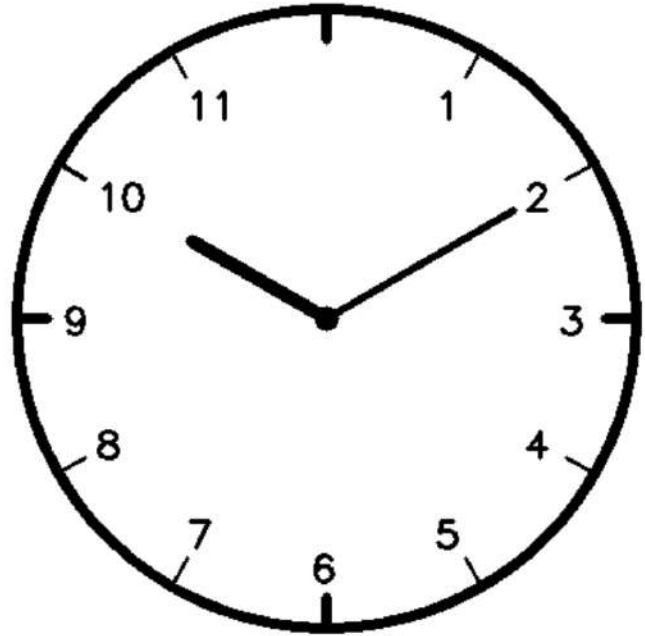
exercise5.clock.py X

```
56
57 # Draw clock hands (showing approximately 10:10)
58
59 # Hour hand at 10
60 hour_angle = math.radians(300 - 90)
61 hour_end = (
62     int(center[0] + 100 * math.cos(hour_angle)),
63     int(center[1] + 100 * math.sin(hour_angle))
64 )
65
66 cv2.line(canvas, center, hour_end, BLACK, 8)
67
68 # Minute hand at 2 (10 minutes)
69 min_angle = math.radians(60 - 90)
70 min_end = (
71     int(center[0] + 140 * math.cos(min_angle)),
72     int(center[1] + 140 * math.sin(min_angle))
73 )
74
75 cv2.line(canvas, center, min_end, BLACK, 4)
76
```

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PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise5.clock.py

Analogue Clock



The image shows a screenshot of a code editor (VS Code) with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named 'PYTHON_TEST' with various files and folders. The file 'exercise5.clock.py' is selected. The code in the editor is a Python script that uses OpenCV to draw an analog clock. The clock is drawn on a canvas with a center point. The hour hand is drawn with a thickness of 8 pixels and the minute hand with a thickness of 4 pixels. The clock face is represented by a circle with tick marks for each hour. The time shown is approximately 10:10. The terminal shows the command 'python exercise5.clock.py' being executed.

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- output.tiff
- pixel_basic C:\Users\raghu\OneDrive\Desktop\python_test\output.tiff
- red.jpg
- saved_image.jpg
- saved_image.png
- target.png

OUTLINE

TIMELINE

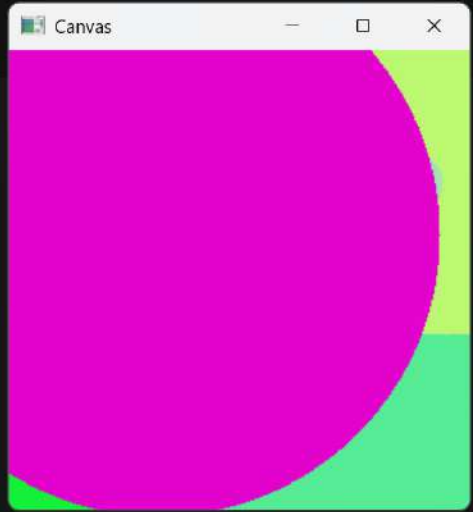
drawing.py X

```
1 # USAGE
2 # python drawing.py
3
4 # Import the necessary packages
5 import numpy as np
6 import cv2
7
8 # Initialize our canvas as a 300x300 image with 3 channels
9 # (Blue, Green, Red) and a black background
10 canvas = np.zeros((300, 300, 3), dtype="uint8")
11
12 # Draw a green line from the top-left corner to the bottom-right
13 green = (0, 255, 0)
14 cv2.line(canvas, (0, 0), (300, 300), green)
15
16 cv2.imshow("Canvas", canvas)
17 cv2.waitKey(0)
18
19 # Draw a 3-pixel thick red line from the top-right corner
20 # to the bottom-left
```

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PS C:\Users\raghu\OneDrive\Desktop\python_test> python drawing.py

Canvas



power...
powersh...
powersh...
powersh...
powersh...
powersh...
powersh...
powersh...
powersh...
powershell
powershell
powershell
powershell
python